

and edit any specific part of the blockchain in an exclusive manner, and no editing can be done by any other entity. As each key is modified within its own domain, it keeps the whole chain coordinated in an efficient manner.

3. **Block creation:** When A starts and locks the transaction, a block is created, which contains encrypted information of the person who intends to transfer the money to B. It is created by combining a timestamp, all significant information as well as a digital signature to ensure that the private key of A is attached to the public key of B to transfer the amount in the form of cryptocurrency.
4. **Noding:** After the secure encrypted block is created, it is broadcast as a message across the blockchain via all nodes attached to the network. Nodes are computers connected to a single network. Once the block is communicated to the system, the nodes are primarily responsible for confirming each entry by solving an arithmetical problem connected to it to verify and validate the block and receive Bitcoins in return.
5. **Verification:** Once the transaction is confirmed, the blocks are time stamped and coupled together with previous verified blocks present in the blockchain. Once the process is complete, Party B receives the amount from Party A.

Open source blockchain platforms

Blockchain technology was first used for financial transactions, but nowadays, its scope is increasing rapidly. Today it is applied in a variety of industries like e-commerce, data management, energy, online voting, gaming, e-governance and many more.

A number of commercial and open source platforms provide the framework for creating applications that support a blockchain. A blockchain product must fulfil the following primary requirements:

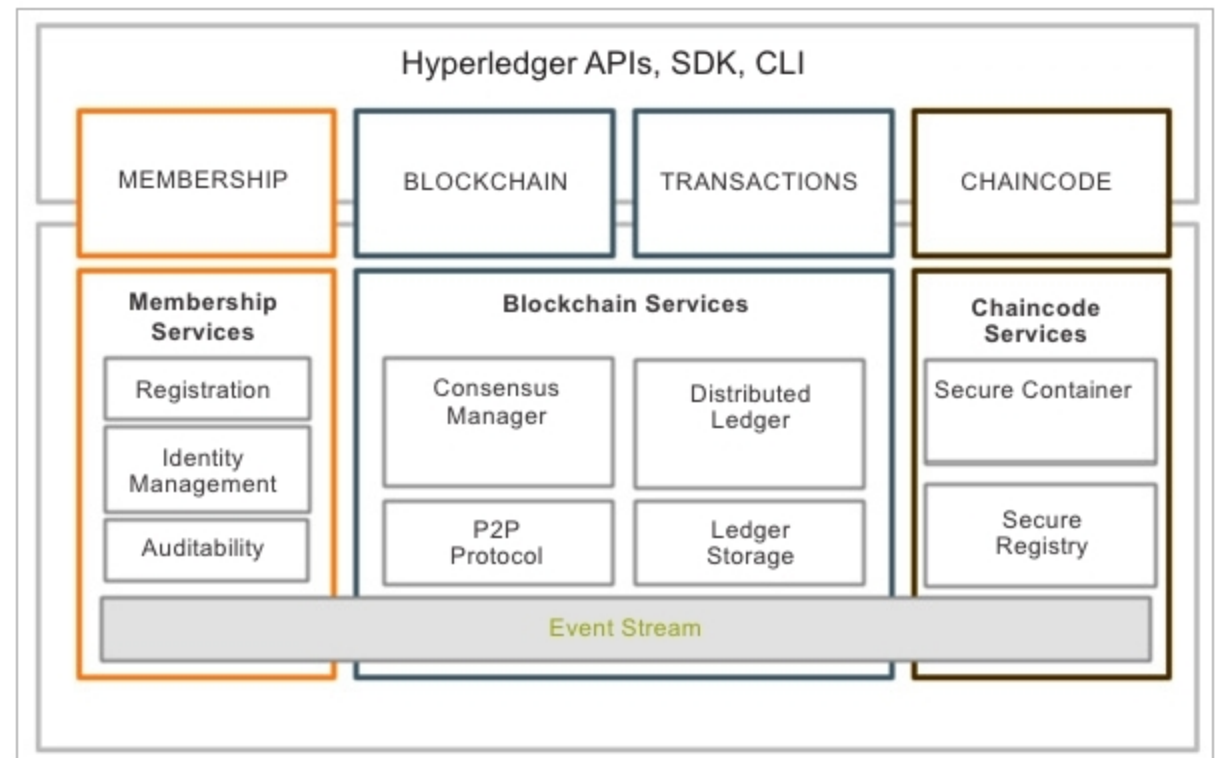


Figure 2: Hyperledger services

- Create a live and distributed transactional database
 - Create user identification labels for active parties
 - Each transaction must be verified on an active ledger for approval
 - Transactions should stop in case of any non-verification
- To develop a smart mobile blockchain app, one needs to use a transaction and ledger model. For easy and quick blockchain app development, there are various open source platforms available, some of which are featured here.

Ethereum

Ethereum is an open source, public, blockchain based decentralised platform running smart contracts — applications that run exactly as programmed without any possibility of downtime, censorship, fraud or third-party interference. It was proposed in 2013 by Vitalik Buterin and was crowdfunded online between July and August 2014.

Ethereum is a distributed public blockchain network and is somewhat different from Bitcoin in terms of capability and purpose. The Bitcoin blockchain is used to track ownership of digital currency, while the Ethereum blockchain mostly focuses on running the programming code of any decentralised application. In the

Ethereum blockchain, Ether is a crypto type which is the heart of the network. Ether is used by application developers to pay transaction fees and services on the Ethereum network.

Ethereum is made up of the following components.

- **Ethereum virtual machine (EVM):** This is a 256-bit register stack that's completely sandboxed and isolated from the network, file system or other processes. Every node runs an EVM and executes the same instructions.
- **Smart contracts:** These are high level programming abstractions compiled to EVM byte code and deployed to the Ethereum blockchain for execution. They are programmed using various languages like Solidity, Serpent, LLL and Mutan.
- **Applications:** Ethereum has almost 50 per cent of the market share and more than 250 live Dapps, which are used for various applications like digital signatures, stock market predictions, gaming, social media, etc.
- **Performance:** It makes use of Merkle trees to improve scalability and optimise transaction hashing.

Official website:

<https://www.ethereum.org/>

Latest version: 0.10.0